P-Polarization Derivation with Wave-numbers

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Conditions

Interface at $z=0$
Plane of incidence is the x-z plane $\Longrightarrow$ $E \in xz$ plane

Electric Field in terms of Wave propogation number

$$E_i = \begin{pmatrix} E_x \\ 0 \\ -\frac{k_x}{k_{z1}} E_x \end{pmatrix} a_p = \begin{pmatrix} \frac{k_{z1}}{k_1} \\ 0 \\ -\frac{k_x}{k_1} \end{pmatrix} a_p$$

$$E_r = \begin{pmatrix} \frac{k_{z1}}{k_1} \\ 0 \\ \frac{k_x}{k_1} \end{pmatrix} r_p$$

$$E_t = \begin{pmatrix} \frac{k_{z2}}{k_2} \\ 0 \\ \frac{-k_{11}}{k_2} \end{pmatrix} t_p$$

Maxwell's Equations for Magnetic Fields

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

Electric field as a plane wave $\longrightarrow E(r,t) = E_0 e^{i(k\cdot r - \omega t)}$

Spacial
$$\frac{\partial}{\partial x} e^{i(k_x x + k_z z - \omega t)} = i k_x x e^{i(k_x x + k_z z - \omega t)}$$
$$\frac{\partial}{\partial z} e^{i(k_x x + k_z z - \omega t)} = i k_z z e^{i(k_x x + k_z z - \omega t)}$$
$$\Longrightarrow \nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \simeq i\left(k_x x, 0, k_z z\right)$$

Temporal
$$\frac{\partial}{\partial t} e^{i(k\cdot r - \omega t)} = -i\omega e^{i(k\cdot r - \omega t)}$$

Plug into Faraday's Law:

$$\nabla \times E = -\frac{\partial B}{\partial t}$$
$$ik \times E = -(-i\omega) B$$
$$ik \times E = i\omega B$$
$$\frac{1}{\omega}(k \times E) = B \quad , \quad B = \mu H$$

$$\therefore H = \frac{1}{\omega \mu} k \times E$$

$$H_i = \frac{1}{\omega \mu}(k_i \times E_i) \Longrightarrow k_i \times E_i = \begin{vmatrix} k_x & 0 & k_{1z} \\ E_x & 0 & E_{iz} \end{vmatrix}$$

$$H_{iy} = \frac{1}{\omega \mu}(k_x E_{iz} - k_{1z} E_{ix}) \qquad = \hat{x}(0) - \hat{y}(k_x E_{iz} - k_{1z} E_{ix}) + \hat{z}(0)$$

Sub in for E_{ix}: $k_{1z} E_{iz} = -k_x E_{ix}$

$$H_{iy} = \frac{1}{\omega \mu}\left[k_{1z} E_{ix} - k_x \left(-\frac{k_x}{k_{z1}} E_{ix}\right)\right]$$

$$H_{iy} = \frac{1}{\omega \mu}\left[k_{1z} E_{ix} + \frac{k_x^2}{k_{1z}} E_{ix}\right]$$

$$H_{iy} = \frac{E_{ix}}{\omega \mu}\left(\frac{k_{1z}^2 + k_x^2}{k_{1z}}\right) \Longleftarrow k_1^2 = k_{1z}^2 + k_x^2$$

$$H_{iy} = \frac{k_1^2}{\omega \mu_1 k_{1z}} E_{ix}$$

Speed of electromagnetic wave in a material: $v = \frac{1}{\sqrt{\upsilon \varepsilon}}$, $v = \frac{\omega}{k}$

$$\frac{\omega}{k} = \frac{1}{\sqrt{\upsilon \varepsilon}}$$

$$k = \omega \sqrt{\upsilon \varepsilon}$$

$$k_1^2 = \omega^2 u_1 \varepsilon_1$$

$$H_{iy} = \frac{\omega^2 \mu_1 \varepsilon_1}{\omega \mu_1 k_{1z}} E_{ix}$$

$$\boxed{H_{iy} = \frac{\omega \varepsilon_1}{k_{1z}} E_{ix}}$$

b/c $k_r = (k_x, 0, -k_{1z})$ & $E_r = (E_{rx}, 0, E_{rz})$
the cross product for H_{ry} is negative.

$$\boxed{H_{ry} = -\frac{\omega \varepsilon_1}{k_{1z}} E_{rx}}$$

For transmitted H_y, k is positive and in medium 2

$$\boxed{H_{ty} = \frac{\omega \varepsilon_2}{k_{2x}} E_{tx}}$$

Apply Boundary Conditions:

$E_{ix} + E_{rx} = E_{tx} \longleftarrow$ B/c the tangential component must be continuous at $z=0$.

$$H_{iy} + H_{ry} = H_{ty}$$

$$\frac{\omega \varepsilon_1}{k_{1z}} E_{ix} - \frac{\omega \varepsilon_1}{k_{1z}} E_{rx} = \frac{\omega \varepsilon_2}{k_{2x}} E_{tx}$$

$$\frac{\varepsilon_1}{k_{1z}}(E_{ix} - E_{rx}) = \frac{\varepsilon_2}{k_{2z}} E_{tx}$$

Reflection Coefficient:

$$r_p = \frac{E_{rx}}{E_{ix}} \qquad t_p = \frac{E_{tx}}{E_{ix}}$$

$$1 + r_p = t_p$$

$$\frac{\varepsilon_1}{k_{1z}}(1 - r_p) = \frac{\varepsilon_2}{k_{2z}} t_p$$

$$\frac{\varepsilon_1}{k_{1z}}(1 - r_p) = \frac{\varepsilon_2}{k_{2z}}(1 + r_p)$$

$$\varepsilon_1 k_{2z} - \varepsilon_1 k_{2z} r_p = \varepsilon_2 k_{1z} + \varepsilon_2 k_{1z} r_p$$

$$-\varepsilon_1 k_{2z} r_p - \varepsilon_2 k_{1z} r_p = \varepsilon_2 k_{1z} - \varepsilon_1 k_{2z}$$

$$-r_p(\varepsilon_1 k_{2z} + \varepsilon_2 k_{1z}) = \varepsilon_2 k_{1z} - \varepsilon_1 k_{2z}$$

$$\boxed{r_p = \frac{-\varepsilon_2 k_{1z} + \varepsilon_1 k_{2z}}{\varepsilon_1 k_{2z} + \varepsilon_2 k_{1z}}}$$